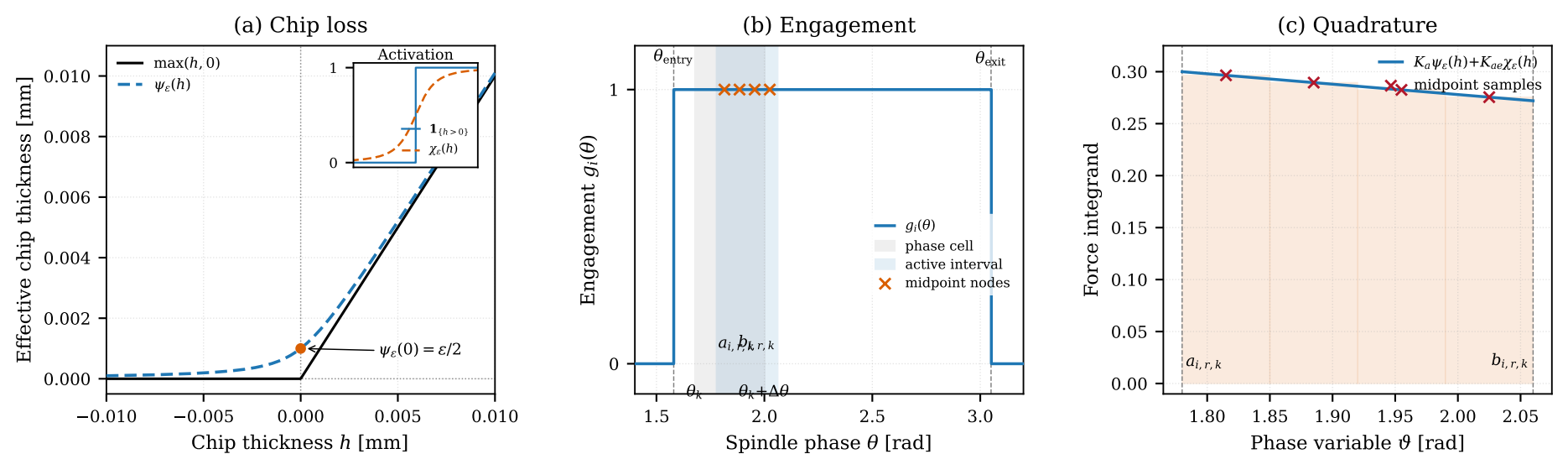
(a) Chip loss
max(h, 0)
ψε(h)
Activation
1
0
1{h>0}
χε(h)
ψε(0) = ε/2
Effective chip thickness [mm]
0.010
0.008
0.006
0.004
0.002
0.000
−0.010
−0.005
0.000
0.005
0.010
Chip thickness h [mm]
(b) Engagement
θentry
θexit
1
0
gi(θ)
phase cell
active interval
midpoint nodes
ai, bi, r, k
θk
θk+Δθ
Engagement gi(θ)
1.5
2.0
2.5
3.0
Spindle phase θ [rad]
(c) Quadrature
Ka ψε(h)+Kae χε(h)
midpoint samples
Force integrand
0.30
0.25
0.20
0.15
0.10
0.05
0.00
ai, r, k
bi, r, k
1.80
1.85
1.90
1.95
2.00
2.05
Phase variable ϑ [rad]